| Qns | Answer | Marks |
|---|---|---|
| **8(a)(i)1** | $\alpha$ particles have <u>greater ionisation power</u> than X-rays, $\gamma$ rays and energetic $\beta$ rays.<br><br>Hence, there is <u>greater damage</u> (biological effect stated in the last sentence in the 2nd last para on pg 18) produced by $\alpha$ particles. | **B1**<br><br><br>**B1** |
| **8(a)(i)2** | Neutrons can be absorbed by the nucleus of an atom to <u>form a radioactive isotope</u>, which is unstable and produces secondary <u>ionising radiations.</u> | **B1** |
| **8(a)(ii)** | **Shield** from source using suitable materials.<br>**Increase the distance** from a source. | **B2** |
| **8(a)(iii)** | Absorbed dose<br>$=\dfrac{\text{energy deposited per year}}{\text{mass of tissue}}$<br>$=\dfrac{(\text{no. of decay})(\text{energy released per decay})}{\text{mass of tissue}}$<br>$=\dfrac{\left[(3.70\times10^{4})(365\times24\times3600)\right]\left[(5.23\times10^{6})(1.60\times10^{-19})\right]}{2.00}$<br>$=0.489\text{ Gy}$<br><br>Dose equivalent<br>$=(\text{absorbed dose})\times\text{RBE}$<br>$=0.489\times20$<br>$=9.8\text{ Sv year}^{-1}$ | <br><br><br><br>**C1**<br>**C1**<br><br><br><br>**A1** |

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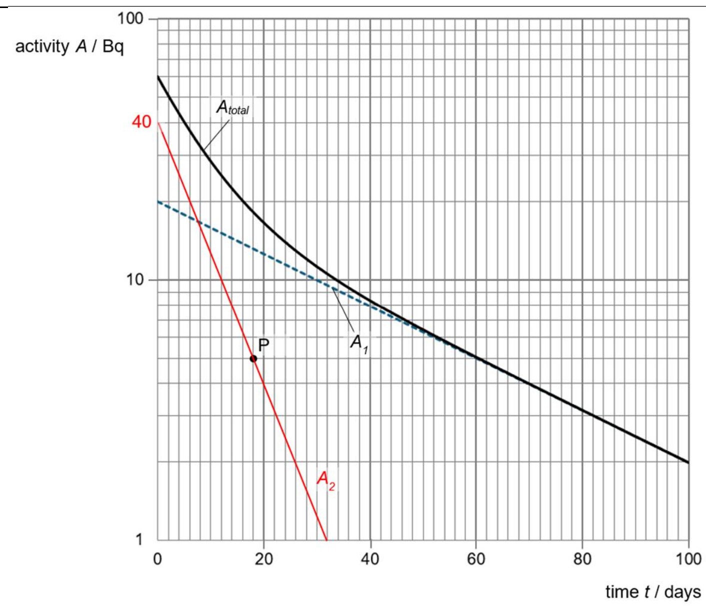
|---|---|---|
| **8(b)(i)** | Gamma rays can pass through the tissue of the body to be captured by the **camera outside** while beta particles will be stopped by the body tissue. | **A1** |
| **8(b)(ii)** | $\lambda_E = \lambda_B + \lambda_T$<br>$\dfrac{\ln 2}{t_E} = \dfrac{\ln 2}{t_B} + \dfrac{\ln 2}{t_T}$<br>$t_E = \dfrac{t_T t_B}{t_T + t_B}$ | **B1** |
| **8(b)(iii)1** | $A_o = \lambda N_o$<br>$= \left(\dfrac{\ln 2}{t_T}\right)\left(\dfrac{M}{\text{molar mass}} \times N_A\right)$<br>$= \left(\dfrac{\ln 2}{6.02 \times 3600}\right)\left(\dfrac{1.0 \times 10^{-12}}{99} \times 6.02 \times 10^{23}\right)$<br>$= 1.94 \times 10^5$<br>$= 1.9 \times 10^5$ Bq | **B1**<br>**B1** |
| **8(b)(iii)2** | $t_E = \dfrac{t_T t_B}{t_T + t_B}$<br>$= \dfrac{6.02 \times 24}{6.02 + 24}$<br>$= 4.8$ h | **A1** |
| **8(b)(iii)3** | $A = \left(\dfrac{1}{2}\right)^{t/t_E} A_o$<br>$= \left(\dfrac{1}{2}\right)^{\frac{3.0 \times 24}{4.8}} \left(1.9 \times 10^5\right)$<br>$= 5.8$ Bq | **C1**<br>**A1** |

| | | |
|---|---|---|
| **8(c)(i)** | After a long enough time, the activity of the nucleus having the shorter half-life would have decreased to a **negligible** value compared to the activity of the nucleus having the longer half-life. Hence the total activity would be the same as the activity of the nucleus having the longer half-life. | **A1** |
| **8(c)(ii)** | From graph, time taken for activity of nucleus 1 to decrease from an initial value of 20 Bq to 10 Bq is 30 days, 10 Bq to 5 Bq is 30 days. Hence average half-life = 30 days | **A1** |
| **8(c)(iii)** | At $t$ = 0 day,<br>$A_{total} = A_1 + A_2$<br>$60 = 20 + A_2$<br>$A_2 = 40\ \text{Bq}$ | **A1** |
| **8(c)(iv)** |  | **B1**<br>• line passes through point P and $y$-intercept at 40 Bq<br>• line must be labelled |
| **8(c)(v)** | From graph, time taken for activity of nucleus 2 to decrease from an initial value of 40 Bq to 20 Bq is 6 days, 20 Bq to 10 Bq is 6 days, 10 Bq to 5 Bq is 6 days. Hence average half-life is 6 days. | **B1** |